

REVIEW

Blood Pressure Variability and Alzheimer Disease Biomarkers

Trevor Lohman¹, Isabel Sibley, Yuan Ma², Niema M. Pahlevan³, Daniel A. Nation¹

ABSTRACT: Increased blood pressure variability (BPV) is associated with increased risk for dementia, independent of hypertension. The increased risk for dementia associated with high BPV includes probable Alzheimer disease (AD) dementia, but high BPV is also associated with increased risk of stroke and cerebral small vessel disease, which are commonly comorbid with AD. This raises the question of whether high BPV is independently predictive of AD-specific pathophysiology and neurodegeneration, which could have significant clinical implications. Fewer studies have directly addressed this question by examining whether higher BPV predicts future AD biomarker abnormalities that are specific to AD pathophysiology within the amyloid/tau/neurodegeneration framework. This narrative review summarizes current studies on this topic and discusses future directions in the investigation of elevated BPV as a risk factor for AD, the most common cause of dementia in older adults.

Key Words: attention ■ blood pressure ■ cerebral small vessel diseases ■ hypertension ■ risk factor

Hypertension has been extensively investigated in relation to risk for Alzheimer disease (AD) and dementia.^{1–3} More recently, attention has shifted to how blood pressure variability (BPV) relates to both AD pathophysiology and clinical dementia syndromes, independent of hypertension. A meta-analysis of data from >7 million older adults found that high BPV is associated with a higher risk for dementia than is mean blood pressure, and that high BPV is specifically linked to an increased risk of probable AD dementia.⁴ This has major implications for the prevention, early detection, and treatment of AD, since elevated BPV could be a highly prevalent therapeutic target in both hypertensive and normotensive older adults.⁵

It has been hypothesized that links between average blood pressure measures and AD pathophysiology may in fact be primarily driven by associated increases in BPV.⁵ However, relatively few studies have tested this hypothesis. Additionally, increased BPV has been linked to a higher risk of stroke⁶ and cerebral small vessel disease,^{7,8} which are commonly comorbid in patients with AD dementia.⁹ To investigate whether elevated BPV is a risk factor for AD-specific pathophysiological change, rather than vascular conditions that are often comorbid

with AD dementia, studies must examine molecular markers of AD as outcomes.

Molecular markers of AD are determined based on the amyloid/tau/neurodegeneration conceptual framework for the diagnosis and staging of AD,¹⁰ which groups biofluid and molecular imaging markers of AD based on their role in AD pathophysiology and the timing of core biomarker abnormality during the AD process.¹¹ In what follows, we summarize and critically review the current studies of BPV and AD amyloid/tau/neurodegeneration framework biomarkers that have been published to date and discuss future directions in the investigation of elevated BPV as a risk factor for AD, the most common neuropathological substrate for dementia in older adults.

BPV Over Time and During Aging

Blood pressure is inherently variable over any timescale, but the determinants of BPV are distinct across these timescales (Figure 1A). For example, beat-to-beat BPV is believed to be predominantly determined by the cardiac and vascular limbs of the baroreceptor reflex,¹² which influence both heart rate variability and

Correspondence to: Daniel A. Nation, Merle H. Bensinger Professor of Gerontology, Leonard Davis School of Gerontology, Professor of Medicine, Keck School of Medicine, University of Southern California. Email danation@usc.edu

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Nonstandard Abbreviations and Acronyms

AD	Alzheimer disease
ADNI	Alzheimer’s Disease Neuroimaging Initiative
Aβ	amyloid-β
BBB	blood-brain barrier
BPV	blood pressure variability
CCB	calcium channel blocker
CSF	cerebrospinal fluid
GFAP	glial fibrillary acidic protein
NfL	neurofilament-light chain
PET	positron emission tomography
ROSMAP	Religious Orders Study and the Rush Memory and Aging Project
SPRINT	Systolic Blood Pressure Intervention Trial

peripheral resistance to modulate blood pressure allostasis.¹³ Over more extended periods of minutes, hours, and days, additional dimensions of BPV include dietary patterns, psychological stress responses, temperature changes, kidney function, nocturnal dipping, morning surge, poor antihypertensive medication adherence,¹⁴ and circadian rhythm.¹⁵ Over months to years, there are

also seasonal variations, changes in lifestyle, medication, and the impact of systemic disease processes.¹⁶ Increasing BPV over years to decades is associated with uncontrolled hypertension, and is also observed with aging,¹⁷ which may be due to a variety of age-related factors such as arterial stiffening¹⁸ and the cardiac and vascular consequences of age-related vagal withdrawal.¹⁹ Thus, the assessment of BPV over different timescales may be of value for specific clinical and research purposes, and BPV over different timescales may show distinct associations and predictive relationships with specific AD pathophysiological changes.

Methods for the quantification of BPV may also vary within and across timescales depending on the number of measures and their temporal resolution (Table 1). The SD and coefficient of variation of multiple blood pressure measurements are simple metrics of BPV. However, both are often correlated with mean blood pressure, which makes it difficult to assess the independent effects of BPV. Variability independent of the mean²⁰ was developed to quantify a statistical parameter of variability that is not correlated with mean blood pressure and is now commonly used in studies of BPV. However, values derived by calculating variability independent of the mean are unique to the derivation cohort and cannot be compared with individuals outside of the derivation cohort.

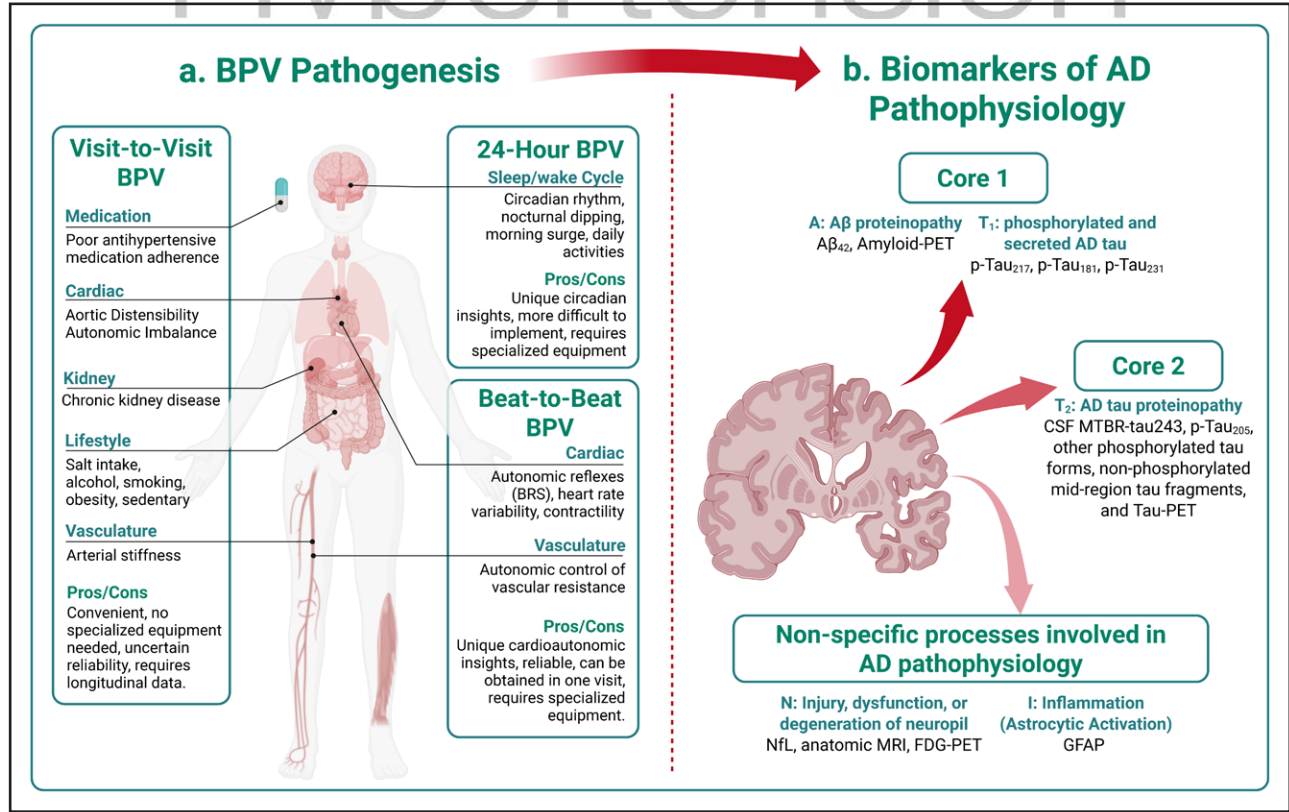


Figure 1. Blood pressure variability (BPV) pathogenesis factors and biomarkers of Alzheimer disease pathophysiology.

Table 1. BPV Metrics

Metric	Characteristics
SD	Simple to calculate, easy to interpret, but highly correlated with mean BP. Average BP should be statistically covaried.
CoV	Generally, less correlated with mean BP
VIM	Less correlated with mean BP but cannot be compared across samples.
ARV	Accounts for the temporal ordering of BP measurements and may represent a reliable high-frequency measure of BPV when used in beat-to-beat analysis.

ARV indicates average real variability; BP, blood pressure; BPV, blood pressure variability; CoV, coefficient of variation; and VIM, variability independent of the mean.

Importantly, none of these measures takes the temporal ordering of blood pressure measurements into account, which may be a relevant hemodynamic characteristic. Average real variability^{21–23} was developed to account for the temporal ordering of blood pressure measures by calculating the average absolute difference between consecutive, time-ordered blood pressure readings. In beat-to-beat analysis, average real variability is shown to be the most reliable measure of BPV, particularly in individuals taking antihypertensive medications, possibly due to decreased susceptibility to outlying values.²⁴ Average real variability can capture beat-to-beat changes in blood pressure and is therefore more sensitive to high-frequency BPV. Average real variability is also less influenced by low-frequency Mayer waves²⁵ which may not be relevant to intrinsic cardiovascular dynamics impacting BPV.

Studies have also examined BPV using systolic, diastolic, mean arterial pressure, and pulse pressure variability. However, the majority of studies have found systolic BPV to be more consistently related to key outcomes.^{4,17,26–31} The present narrative review focuses specifically on findings relating systolic BPV to AD molecular marker outcomes.

AD Pathophysiology and Its Molecular Markers

The dominant model for AD pathophysiology is the amyloid cascade hypothesis, which states that the cerebral retention of extracellular Aβ (amyloid-β) protein in the form of toxic oligomers and insoluble plaques triggers abnormal hyperphosphorylation of the microtubule-associated protein tau, leading to the formation of intracellular neurofibrillary tangles comprised of paired helical filaments of abnormal tau protein.³² The combination of these 2 protein pathologies results in the spread of abnormal tau seeds throughout the cerebral neocortex and limbic cortex,^{33,34} leading to synapse loss and neurodegeneration that drives cognitive decline, dementia, and ultimately death. With the advent of molecular markers that detect cerebral Aβ retention and tau accumulation during life, a prolonged

asymptomatic and prodromal period has been identified.³⁵ During this prodromal disease phase, the AD process is underway, but little or no clinical symptoms are apparent.

The continued development and testing of novel blood- and cerebrospinal fluid (CSF)-based biomarkers of Aβ and tau proteins, as well as radioligands for positron emission tomography (PET) molecular imaging of Aβ and tau pathology, have improved the ability to track the AD pathophysiological process at an early stage. A recent revision to the criteria for the diagnosis and staging of AD has defined the biomarkers of AD pathophysiology based on their ability to detect AD-specific and nonspecific processes at various stages¹¹ (Figure 1B).

Core 1 markers represent the earliest detectable biomarkers indicative of AD pathophysiological changes. These include markers of Aβ proteinopathy (core 1 A) and the earliest detectable changes in AD tau proteins (core 1 T₁). The core 1 A markers of Aβ proteinopathy are abnormally low levels of Aβ₄₂ in the CSF or blood plasma, indicative of increased cerebral Aβ retention. Cerebral Aβ plaques may also be detected at this early stage using amyloid-PET imaging. The core 1 T₁ markers of AD tau proteinopathy include a set of abnormally phosphorylated tau (pTau) proteins detectable at an early stage in CSF or blood plasma³⁶ (pTau₂₁₇, pTau₁₈₁, pTau₂₃₁). It should also be noted that ratios are often utilized, particularly in studies using blood plasma markers, to better capture Aβ and tau changes specific to AD (eg, plasma Aβ_{42/40} ratio, CSF pTau₁₈₁/Aβ₄₂).

Core 2 T₂ markers represent markers of AD tau proteinopathy that are detectable at a later stage and include abnormally increased CSF and blood plasma levels of microtubule tau binding region-tau₂₄₃,³⁷ pTau₂₀₅,³⁸ and nonphosphorylated mid-region tau fragments,³⁹ as well as detection of tau accumulation on tau-PET imaging. For tau-PET imaging, tau accumulation is typically first detectable in the temporal lobe, and at a later stage across other neocortical regions.

Additional biofluid and imaging markers are also recognized as biomarkers of nonspecific processes that are important in AD pathophysiology and staging.¹¹ These include biomarkers of injury, dysfunction, or degeneration of the neuropil (N), including increased CSF or blood plasma NfL (neurofilament light), AD pattern volume loss on brain magnetic resonance imaging, and AD pattern hypometabolism on fluorodeoxyglucose-PET imaging. The level of total tau in CSF and blood plasma has also been considered to be a biomarker of neurodegeneration in previous research recommendations for the diagnosis of AD.^{10,40–42} However, total tau levels are less specific to AD and can be impacted by peripheral factors when measured in blood plasma.

Another biomarker of nonspecific processes involved in AD pathophysiology is GFAP (glial fibrillary acidic protein), which is a biomarker of astrocyte activation

and inflammation (I)^{43,44} that closely tracks AD pathophysiology,^{43–50} specifically when measured in plasma, but may also be associated with other neurological disorders.^{51–53} Lastly, the amyloid/tau/neurodegeneration framework includes biomarkers of common AD copathologies, such as vascular brain injury (V). Although there has been debate over whether certain vascular pathologies may represent key components of AD pathophysiology, rather than co-pathologies,^{54–57} the current review focuses specifically on the current AD biomarker framework from the most recent diagnostic recommendations.¹¹

Although genetics was not included in the most recent revisions to the diagnosis and staging of AD, the apolipoprotein E4 (*APOE4*) gene is a prevalent (15%–25% of the population) risk gene for AD that is associated with a 2–4-fold risk for AD in heterozygotes and 10- to 15-fold risk for AD in homozygotes.⁵⁸ Importantly, *APOE4* is also associated with increased risk for vascular disease and contributes to cognitive decline via both vascular dysfunction and cerebral amyloidosis through distinct mechanisms.^{58,59}

BPV and AD Biomarkers

Here, we review studies that have investigated BPV at multiple timescales, using different methods for quantification of BPV, in relation to different AD-specific core biomarkers (Core 1 A, T₁, and T₂ markers) and biomarkers of nonspecific processes considered to be important in AD pathophysiology (N and I) in both cross-sectional and longitudinal study designs. Table 2 displays details of studies and cohorts in which the relationship between BPV and AD biomarkers has been investigated.^{28,60–69}

Visit-to-Visit BPV and Core 1 AD Biomarkers

The AD pathophysiological process is insidious in its onset and slowly progressive in its course, with a years-long prodromal period during which AD biomarker abnormality may be observed in the absence of any symptoms. These features of the AD process make prospective, longitudinal studies logistically challenging. Due to these challenges, most of the relevant BPV studies to date are retrospective in nature, using visit-to-visit BPV as a convenience measure that allows for the leveraging of



Table 2. Findings and Cohort Characteristics of Studies of Blood Pressure Variability and Alzheimer Disease Biomarkers

Visit-to-visit blood pressure variability						
Study title	Outcome	BPV metric	Biomarker category	Longitudinal (yes/no)	Cohort	Sample size
Sible et al ⁶⁰	↑ Tau-PET. Greatest in <i>APOE4</i> carriers.	VIM	T ₂	Yes	ADNI	286 (100 <i>APOE4</i> carriers)
Sible et al ⁶¹	↓ CSF Aβ ₄₂ , ↑ CSF total tau, ↑ CSF p-Tau ₁₈₁ . p-Tau ₁₈₁ effect moderated by <i>APOE4</i> carrier status.	VIM	A, T ₁	Yes	ADNI	466 (148 <i>APOE4</i> carriers)
Lee et al ²⁸	↑ Tau-PET	SD	T ₂	No	Samsung Medical Center Memory Clinic	106
Li et al ⁶²	↑ Plasma NfL	SBP VIM, DBP VIM, PP VIM, MAP VIM	N	No	ADNI	413
Li et al ⁶³	↑ Aβ-PET	SD	A	No	ADNI	658
Sible et al ⁶⁴	↑ Plasma total tau, null finding: Aβ ₄₂ /Aβ ₄₀ ratio	VIM	A	Yes	SPRINT	457
24-Hour ambulatory blood pressure variability						
Starmans et al ⁶⁵	Null findings: plasma NfL, plasma p-Tau ₁₈₁ , plasma Aβ ₄₂ , plasma Aβ ₄₀ , plasma Aβ _{42/40} ratio.	24-h ARV, Night ARV, day ARV. Nocturnal dipping ratio	A, T ₁ , N	No	Heart-Brain Connection Study	156
Zou et al ⁶⁶	↓ CSF Aβ _{42/40} ratio, ↑ CSF total tau	CoV	A	No	CIBL	184
Beat-to-beat blood pressure variability						
Sible et al ⁶⁷	↓ Plasma Aβ ₄₂ , ↓ plasma Aβ ₄₂ /Aβ ₄₀ ratio, ↑ plasma total tau, ↑ plasma p-Tau ₁₈₁ . Null findings: p-Tau ₁₈₁ ratio.	VIM	A, T ₁	No	VaSC	54
Lohman et al ⁶⁸	↑ Plasma GFAP, ↑ plasma NfL	ARV	I, N	No	VaSC	56
Lohman et al ⁶⁹	↑ Plasma GFAP, ↑ plasma NfL	ARV	I, N	No	VaSC	52

A indicates Aβ proteinopathy; ADNI, Alzheimer's Disease Neuroimaging Initiative; *APOE4*, apolipoprotein-ε4; ARV, average real variability; BPV, blood pressure variability; CIBL, Chinese imaging, biomarkers, and lifestyle cohort; CoV, coefficient of variation; CSF, cerebrospinal fluid; DBP, diastolic blood pressure; GFAP, glial fibrillary acidic protein; I, inflammation; MAP, mean arterial pressure; N, injury, dysfunction, or degeneration of neuropil; NfL, neurofilament-light chain; PET, positron emission tomography; PP, pulse pressure; SBP, systolic blood pressure; SPRINT, Systolic Blood Pressure Intervention Trial; T₁, Phosphorylated and secreted Alzheimer disease tau; T₂, Alzheimer disease tau proteinopathy; VaSC, vascular senescence and cognition study; and VIM, variability independent of the mean.

existing longitudinal data. Visit-to-visit BPV also has the advantage of direct clinical applicability, as serial blood pressures are routinely measured during clinical follow-up across a wide variety of patient care settings.

The first studies of BPV in relation to AD pathophysiological markers were conducted through retrospective analysis of data from the ADNI (Alzheimer's Disease Neuroimaging Initiative) cohort, a longitudinal study focused on AD biomarkers in cognitively unimpaired and mildly impaired older adults.⁷⁰ The first BPV study in ADNI compared cognitively unimpaired control participants with those diagnosed with MCI or dementia due to AD, confirmed by abnormal Core 1 CSF markers of both $A\beta_{42}$ (A) and ptau₁₈₁ (T₁). Findings indicated that older adults with MCI or dementia due to AD exhibited elevated visit-to-visit BPV.⁷¹ These findings are consistent with subsequent community-based studies, including data from the ROSMAP (Religious Orders Study and the Rush Memory and Aging Project), which found that higher antemortem visit-to-visit BPV over 9 to 12 years predicted future development of autopsy-confirmed AD dementia.²⁹ Importantly, this study included an additional subanalysis confirming that BPV also predicted pure AD dementia, with no other contributing pathologies being found in the patient's brains at autopsy. Since most patients with AD brains have both AD and vascular pathology, as well as other copathologies, this finding clarifies that elevated BPV does not merely predict comorbid vascular pathology, but also predicts AD-specific pathology in the absence of any other copathologies.

In the first study of BPV predicting Core 1 AD molecular marker changes during life, a longitudinal analysis of ADNI data found that visit-to-visit BPV predicted future longitudinal decreases in CSF $A\beta_{42}$, indicating increased cerebral $A\beta$ proteinopathy. It also predicted increases in CSF pTau₁₈₁,⁶¹ indicating progressive AD tau proteinopathy. Visit-to-visit BPV was predictive of faster longitudinal increases in CSF pTau₁₈₁ in *APOE4* carriers in particular. Findings relating higher visit-to-visit BPV to elevated Core 1 A markers of $A\beta$ proteinopathy were later replicated using amyloid-PET imaging in the ADNI cohort,⁶³ and the link between increased visit-to-visit BPV and longitudinal increases in the Core 1 T₁ biomarkers of pTau₁₈₁ was confirmed in a later neuropathological study, indicating antemortem visit-to-visit BPV predicts greater tau neurofibrillary tangle burden.⁷² Together, these studies provide clear evidence of a link between high visit-to-visit BPV and AD tau proteinopathy, with confirmation by multiple longitudinal studies showing biomarker and neuropathological confirmation across multiple cohorts. However, not all studies linking visit-to-visit BPV specifically to $A\beta$ proteinopathy have been consistent. One retrospective study in the SPRINT clinical trial cohort did not identify any relationship between visit-to-visit BPV and plasma $A\beta_{42/40}$ ratio in hypertensive older adults undergoing standard or intensive blood pressure lowering.⁶⁴

Neuropathological studies have also reported no association between antemortem visit-to-visit BPV and amyloid plaque burden at autopsy, despite finding links with tau tangle burden.⁷² Thus, visit-to-visit BPV studies suggest the link between BPV and AD may be due to effects that predominantly involve AD tau pathophysiology.

Visit-to-Visit BPV and Core 2 AD Biomarkers

In the ADNI cohort, higher visit-to-visit BPV predicted future longitudinal increases in the Core 2 T₂ biomarker of tau proteinopathy on tau-PET imaging in the temporal lobe.⁶⁰ This finding was confirmed in a cross-sectional study of a separate sample of 106 Asian individuals without dementia, in which higher visit-to-visit BPV correlated with increased tau proteinopathy on tau-PET imaging.²⁸ These tau-PET studies are consistent with findings from T₁ markers of tau hyperphosphorylation in CSF⁶¹ and blood,⁶⁷ as well as neuropathological studies correlating antemortem BPV with postmortem burden of neurofibrillary tau tangles.⁷² Thus, visit-to-visit BPV has been consistently linked to both Core 1 T₁ and Core 2 T₂ biomarkers, underscoring a likely role in accelerating AD tau pathology across multiple stages of AD biomarker abnormality.



Visit-to-Visit BPV and Biomarkers of Nonspecific Processes Involved in AD Pathophysiology

Visit-to-visit BPV is associated with plasma NfL,⁶² a biomarker of neural injury, neural dysfunction, and neurodegeneration. Analogous to NfL, total tau levels in CSF or blood plasma may represent a nonspecific marker of neurodegeneration.^{10,11,40–42} Higher visit-to-visit BPV predicted future longitudinal increases in CSF total tau in the ADNI cohort,⁶¹ as well as plasma total tau in hypertensive older adults from the SPRINT cohort.⁶⁴ Longitudinal ADNI data also demonstrate that visit-to-visit BPV predicts future, longitudinal AD-pattern atrophy in medial temporal brain regions susceptible to early-stage AD tau pathology.⁷³ Together with findings linking visit-to-visit BPV to tau pathophysiology, these studies further implicate visit-to-visit BPV in the tau-mediated neurodegeneration that underlies cognitive decline in AD.

Beat-to-Beat BPV and Core AD Biomarkers

Fewer studies have prospectively examined AD biomarker levels associated with BPV across other timescales, including beat-to-beat BPV. Only 1 relatively small study has examined beat-to-beat BPV in relation to Core 1 AD biomarkers. Findings indicated higher beat-to-beat BPV correlated cross-sectionally with lower plasma $A\beta_{42/40}$ ratio and higher plasma pTau₁₈₁/ $A\beta_{42}$,⁶⁷ indicating both cerebral $A\beta$ and tau proteinopathy. There have been no studies relating beat-to-beat BPV to Core 2 AD biomarkers. Additional larger longitudinal studies are needed to clarify whether beat-to-beat BPV predicts AD pathophysiological change, as observed in retrospective visit-to-visit BPV studies. These investigations may have

translational significance since beat-to-beat BPV may be more related to intrinsic vascular physiology and could be a more suitable therapeutic target for intervention studies relative to visit-to-visit BPV.

Beat-to-Beat BPV and Biomarkers of Nonspecific Processes Involved in AD Pathophysiology

Beat-to-beat BPV is associated with plasma NfL.^{68,69} In the VaSC study cohort of community-dwelling older adults, beat-to-beat BPV was also correlated cross-sectionally with higher plasma total tau levels.⁶⁷ Only 2 studies have evaluated the relationship between BPV and biomarkers of astrocyte activation, both of which utilized beat-to-beat BPV and both were conducted within the same cohort. These studies found that increased beat-to-beat BPV correlates cross-sectionally with higher plasma GFAP levels, indicating astrocyte activation and inflammation.^{68,69} There have been no studies to date on visit-to-visit or 24-hour BPV relating to the plasma GFAP levels, underscoring the need for further studies.

24-Hour Ambulatory BPV

To date, there have been only 2 studies of 24-hour ambulatory BPV in relation to AD biomarkers. One study found that higher 24-hour ambulatory BPV was related to lower CSF A $\beta_{42/40}$ ratio, indicating increased A β proteinopathy.⁶⁶ The second study was in a distinct cohort that combined patients diagnosed with vascular cognitive impairment, carotid occlusive disease, heart failure, and control participants. In this study, 24-hour ambulatory BPV was not associated with plasma A $\beta_{42/40}$ ratio, pTau₁₈₁, or NfL levels.⁶⁵ Given the small number of 24-hour ambulatory BPV studies with contrasting results, it remains unclear whether 24-hour BPV has any relationship with AD pathophysiology, but further investigation is warranted.

Summary

The relationship between higher BPV and Core AD biomarkers has been shown across multiple research groups and older adult cohorts. The relationship between elevated BPV and tau proteinopathy markers has been the most consistent finding. Specifically, elevated BPV has been found to be associated with tau proteinopathy in 4 separate cohorts investigated by 3 different research groups. These studies used multiple modalities to assess tau proteinopathy, including blood plasma,⁶⁷ CSF,⁶¹ PET,⁶⁰ and postmortem⁷² methods, and examined both visit-to-visit^{60,61} and beat-to-beat BPV.⁶⁷

The only study reporting null findings for the relationship between BPV and biomarkers of tau proteinopathy was methodologically distinct in its use of 24-hour ambulatory BPV and its focus on individuals with major risk factors along the heart-brain axis.⁶⁵ The cohort included individuals with vascular cognitive impairment, carotid occlusive disease, and heart failure,

and was combined with control participants. In this study, there were no significant associations between 24-hour ambulatory BPV and any of the AD biomarkers studied (plasma A $\beta_{42/40}$, pTau₁₈₁, NfL). It is possible that the predominant inclusion of individuals at risk for or diagnosed with vascular cognitive impairment may have played a role in the null findings relating BPV to AD pathophysiological markers.

It is also possible that 24-hour variability in blood pressure is fundamentally distinct from other forms of BPV since some variation in blood pressure is necessary or even beneficial. For example, nocturnal dipping in systolic blood pressure is protective against dementia.⁷⁴ Additionally, increased physical activity causes a change in blood pressure, yet increased physical activity is associated with decreased dementia risk.⁷⁵ However, at least 1 study has found that increased 24-hour BPV was associated with increased dementia risk.⁷⁶ Future studies of ambulatory BPV should account for nocturnal dipping and BPV associated with physical activity level.

Interestingly, the only 2 longitudinal studies of BPV predicting future longitudinal increases in AD biomarker abnormalities were done in the ADNI study cohort, a cohort with relatively low vascular risk that was specifically designed to target individuals at risk for AD.⁶¹ In these ADNI studies,^{61–63} the links between visit-to-visit BPV and AD biomarker abnormality were also predominantly driven by participants carrying one or more copies of the AD risk gene, *APOE4*. Together, these findings may suggest that elevated BPV is more likely to be related to AD pathophysiological changes in individuals at risk for AD, particularly those carrying *APOE4*. This is consistent with prior studies suggesting vascular risk factors are more strongly linked to cognitive decline in *APOE4* carriers relative to noncarriers.⁷⁷ Further longitudinal studies in distinct cohorts with *APOE* genotype characterization are needed to confirm this possibility.

Findings linking higher BPV to biomarkers of A β proteinopathy were more mixed than those for tau proteinopathy markers.^{28,60,61,64,67} This is consistent with prior work in the ADNI cohort and other study cohorts showing that hypertension and increased systolic blood pressure and pulse pressure are more strongly and consistently related to tau proteinopathy rather than A β proteinopathy.^{78–82} This is also consistent with studies demonstrating that total vascular risk factor burden works synergistically with A β to increase tau accumulation on tau-PET,⁸³ rather than contributing primarily to cerebral A β retention. Age-related arterial stiffening is thought to be a primary driver of increased BPV with aging,¹⁸ and findings from the Framingham Heart Study show that both aortic stiffness and pressure pulsatility are associated with tau accumulation on tau-PET, but not with A β proteinopathy on amyloid-PET.⁸⁴ If links

between BPV and AD pathophysiology are causal in nature, as suggested by Mendelian randomization studies,⁸⁵ it may be hypothesized that BPV particularly impacts cellular and molecular pathways involved in tau hyperphosphorylation.

Potential Mechanisms Linking BPV to AD Pathophysiology

Cerebral hypoperfusion could represent a key mechanism linking high BPV to AD tau accumulation.^{56,86,87} The first studies of BPV in the ADNI cohort showed that visit-to-visit BPV correlated with both decreased cerebral blood flow and brain atrophy over time in AD signature brain regions, including the hippocampus and entorhinal cortex.^{73,88} Cross-sectional studies of beat-to-beat BPV also show a similar pattern of increased BPV correlating with reduced hippocampal and entorhinal perfusion.^{88,89} In the SPRINT cohort, higher visit-to-visit BPV predicts future longitudinal decreases in cerebral blood flow, specifically in the hippocampus and entorhinal cortex.⁹⁰ Given the temporal and anatomic correspondence between hippocampal and entorhinal hypoperfusion, tau deposition, and atrophy, it is possible that BPV is related to AD pathophysiological changes through a cerebral perfusion deficit.⁵⁶ Neuronal hypoxic-ischemic stress can accelerate the accumulation of hyperphosphorylated tau protein,^{86,87,91} and brain regions prone to cerebral hypoperfusion exhibit increased tau neurofibrillary tangle pathology.⁹²

However, it is also possible that the cerebral hypoperfusion observed in association with increased BPV is a secondary result of neurodegeneration, rather than a causal factor. Cerebral hypoperfusion often follows as a consequence of hypometabolism due to neurodegeneration,⁹³ leaving it unclear whether decreased cerebral blood flow is a cause or a consequence of neurodegeneration. This question has been addressed in part through studies of potential cerebrovascular mechanisms of brain injury in older adults with elevated BPV. For example, older adults with increased beat-to-beat BPV exhibit deficits in cerebrovascular vasodilatory reactivity to hypercapnia.⁹⁴ These findings suggest that high BPV correlates not only with resting perfusion, a proxy for neurodegeneration, but also with cerebrovascular-specific functional deficits.

Experimental studies in animal models of BPV also indicate that chronically high BPV leads to deficits in cerebral vasomotor function.⁹⁵ This is consistent with human studies demonstrating that higher BPV correlates with deficits in cerebral vasomotor function.^{94,96} Vasomotion plays a key role in the clearance of toxins and brain waste products, including A β and tau, through the intramural periarterial drainage pathway, and through CSF waste clearance along perivascular spaces.^{97,98} Acute increases in blood pressure caused by angiotensin II infusion disrupt CSF flow through perivascular spaces

in animal models,⁹⁹ providing further evidence that acute increases in blood pressure, similar to those experienced during high BPV, could intermittently disrupt perivascular pathways important for brain waste clearance.

Glymphatic clearance represents another vessel-related pathway for the clearance of AD pathological proteins.¹⁰⁰ Glymphatic clearance of A β and tau relies on the expression of aquaporin-4 water channels on astrocytic end feet.^{101,102} If chronically high BPV impairs cerebral vasomotor function, perivascular CSF flow, and glymphatic clearance, this could lead to decreased clearance of both A β and tau, potentially contributing to AD pathophysiology.

Another distinct vascular mechanism that has been linked to tau pathophysiology is the depletion of nitric oxide bioavailability, which triggers tau hyperphosphorylation through a mechanism involving reduced neural calpain nitrosylation and activation of cyclin-dependent kinase-5.¹⁰³ Reduced nitric oxide bioavailability could represent a final common pathway by which vascular risk factors, including high BPV, contribute directly to AD tau pathophysiology. Recent work has also identified numerous endothelial genes linked to AD,¹⁰⁴ and experimental studies show that peripheral factors, which may include hemodynamic factors like BPV, may drive AD pathophysiological changes in the brain through their effects on endothelial gene expression patterns.^{105–107}

Another intriguing possibility is that astrocyte activation at the neurovascular unit may play a key role in linking high BPV to tau pathology. Astrocyte activation has recently been suggested to serve as a key player in the mechanistic link between A β and tau pathophysiology,¹⁰⁸ and observational studies in aged humans have identified correlations between elevated beat-to-beat BPV and astrocyte activation (plasma GFAP).^{68,69,109} Given the role of astrocytes as sensors in the local autoregulation of parenchymal arteriolar pressure and flow,¹¹⁰ it is possible that high BPV may affect astrocyte physiology. Astrocytes at the neurovascular unit also play a critical role in the glymphatic pathway that is thought to contribute to the clearance of toxic waste products, possibly including A β and tau.^{111,112}

Chronically high BPV could also disrupt the blood-brain barrier (BBB) through mechanical stimuli (eg, shear stress, cyclic strain), which could directly cause injury to the endothelium or alter cell signaling pathways with downstream effects on the brain.^{105,107} Increased BBB permeability has been linked to AD,^{113,114} and specifically with tau pathophysiology.¹¹⁵ Hypertension is known to impair the BBB,^{116,117} and it has been widely hypothesized that the high BPV associated with hypertension is likely the causal mechanism underlying the effects of hypertension on the BBB due to increased shear stress and cyclic strain.¹⁰⁵

Importantly, longitudinal studies find that the predictive relationship between high BPV and AD biomarkers

is predominantly observed in *APOE4* carriers.^{60,61} This could be related to the vulnerability of *APOE4* carriers to neurovascular dysfunction,⁵⁸ and specifically increased BBB permeability.¹¹⁸ To investigate this possibility, future studies should determine whether high BPV predicts BBB permeability, particularly in *APOE4* carriers.

In summary, there are multiple causal mechanisms by which chronically elevated BPV may contribute to the core pathophysiological changes in A β and tau that represent the hallmarks of AD (see Figure 2). In addition, BPV may also act through other mechanisms related to AD pathophysiology, such as astrocyte activation and neurodegeneration. However, these are mostly speculative possibilities, and more studies in humans and experimental models are needed to further establish the specific causal mechanism(s) linking BPV to AD.

Human longitudinal cohort studies^{60,64,71,73,88,90,119} and mendelian randomization studies,⁸⁵ support a causal model for high BPV contributing to AD pathophysiological changes, particularly involving increased AD tau proteinopathy, through a cerebrovascular mechanism. Prior studies of hypertension and other measures of average blood pressure elevation have also demonstrated similar causal links to AD pathophysiology.^{95,102,120} However, it may be hypothesized that links between hypertension and AD are in fact due to increased BPV associated with hypertension. In support of this hypothesis, all studies relating high BPV to AD molecular markers statistically

control for average blood pressure, whereas studies of hypertension or standard blood pressure averages do not control for BPV.

This question might be further addressed by experimental studies in animal models of AD. Experimental studies of hypertension in animal models support a causal model for hypertension, accelerating AD pathophysiological changes.^{95,102,120} To date, there have been few experimental studies in animal models specifically focusing on BPV, highlighting a key gap between basic and translational studies of BPV and AD pathophysiology.⁸⁰ Future studies should aim to disentangle the relative contribution of BPV, versus average blood pressure, to AD pathophysiological changes in both experimental animal models of AD and in human longitudinal and experimental studies.

Lastly, we cannot rule out the possibility that AD-related neurodegeneration could also cause increased BPV. Several brain regions affected relatively early in the course of AD are known to play major roles in higher-order, cortical-level regulation of circulatory function through the cerebral autonomic network.¹²¹ It has been hypothesized that neurodegeneration in these brain regions could cause a dysregulation of autonomic mechanisms critical for BPV regulation.¹²² This potential bidirectional relationship between BPV and AD could represent a vicious cycle that further accelerates neurodegeneration in older adults with BPV elevation.^{69,123}

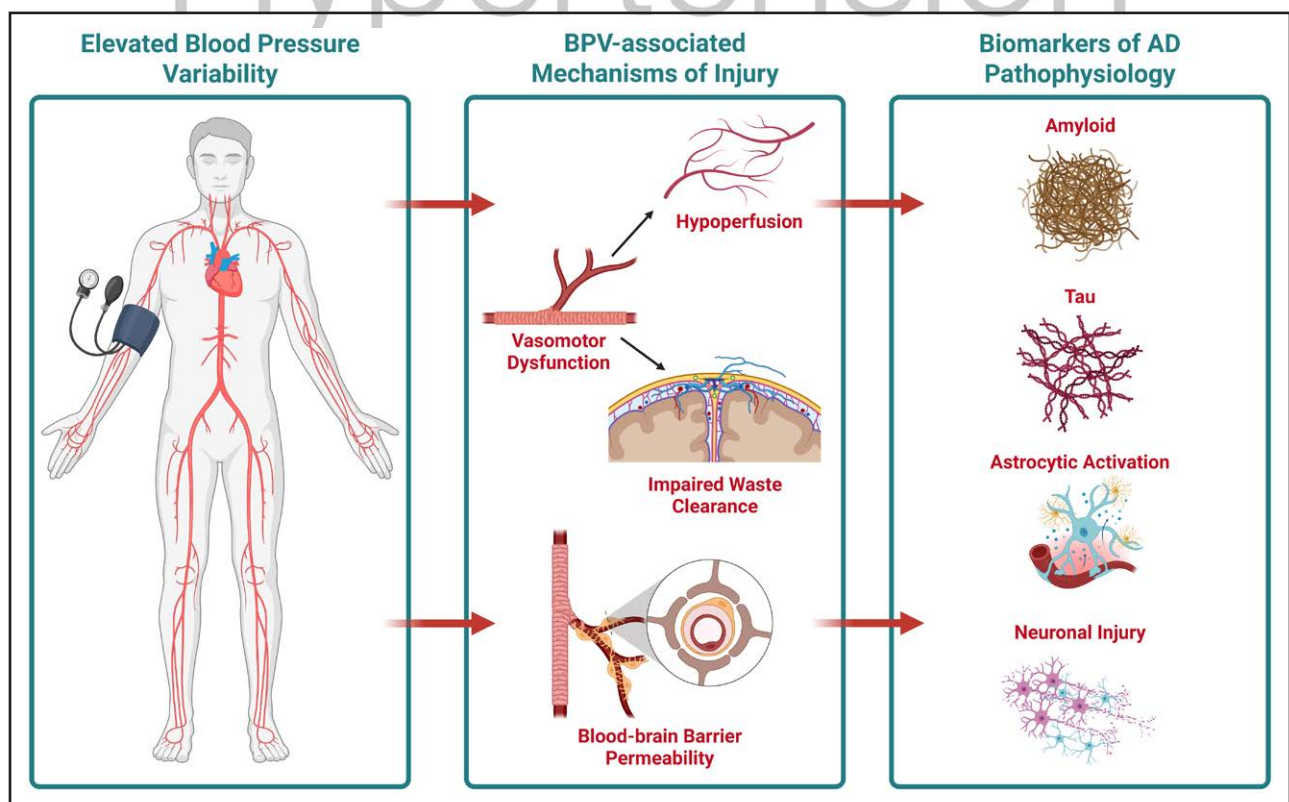


Figure 2. Potential mechanisms of BPV-associated Alzheimer disease pathophysiology.

There is very little direct evidence for a potential vicious cycle between high BPV and AD pathophysiology, but one study did find that decreased connectivity within the central autonomic network correlates with lower plasma $A\beta_{42/40}$ ratio and higher GFAP in nondemented older adults,¹²⁴ suggesting that at an early stage, there is neuronal dysfunction in brain networks potentially impacting BPV. Importantly, higher BPV remains directly correlated with neurodegeneration and memory dysfunction after accounting for any decrease in cerebral autonomic network connectivity.⁶⁹ Longitudinal studies to date have used a cross-lagged design wherein high BPV preceded and predicted future longitudinal increases in all AD biomarkers.^{60,61} To our knowledge, there are no studies describing increased BPV in animal models of AD. Thus, reverse causation is not likely to be responsible for the relationship between BPV and AD.

Treatment Implications

There is no consensus on specific treatments to manage BPV. However, the use of CCBs (calcium channel blockers),^{125–127} particularly long-acting dihydropyridines like amlodipine,¹²⁸ are associated with reduced visit-to-visit BPV and may also diminish dementia risk and $A\beta_{42}$ production.^{129,130} In the community-based ROSMAP study of autopsy-confirmed AD, visit-to-visit BPV was lower in participants taking CCBs, and the link between BPV and future dementia was attenuated in those taking CCBs.²⁹ Similar but nonsignificant trends were observed for renin-angiotensin system drugs in the same study, and another study identified beneficial effects of ARBs on the adverse vascular effects of increased BPV.¹³¹ Further studies are needed to identify the role of antihypertensive medication class, particularly CCBs and renin-angiotensin system drugs, and pharmacokinetics (eg, half-life and BBB penetrance^{132–134}) in lowering BPV and attenuating its associated risk for AD dementia.

Poor antihypertensive medication adherence has also been proposed as a modifiable factor contributing to BPV pathogenesis.^{14,135,136} A post hoc analysis of the SPRINT trial revealed that poor medication adherence was associated with both elevated blood pressure and elevated BPV, and elevated BPV partially accounted for the relationship between suboptimal medication adherence and risk for cognitive impairment.¹⁴ In the ROSMAP study, the link between visit-to-visit BPV and future dementia was attenuated in participants who consistently took their antihypertensive medications relative to those who took their medications less consistently.²⁹ Together, these studies underscore the importance of medication adherence in lower BPV and BPV-associated risk for dementia.

It should be noted that high BPV has been associated with increased risk for dementia and AD biomarker changes in older adults who do not have hypertension based on average blood pressure levels, as well as

hypertensive individuals receiving standard treatment for hypertension.²⁶ This may reflect the fact that high BPV can be observed even when average blood pressure is within normal limits, whether treated or untreated. A major opportunity for the prevention of dementia and AD could be exploited if high BPV could be treated in individuals who are normotensive, and in those who have well-controlled hypertension based on average blood pressure levels. Aggressive lowering of average blood pressure (systolic <120 mm Hg) might represent 1 strategy for lowering BPV in people who are normotensive or have well-controlled average blood pressure under standard care.¹³⁷ However, analysis of data from the SPRINT study indicates that high BPV predicts future longitudinal decline in hippocampal perfusion and increased risk for cognitive impairment and dementia, even when average blood pressure levels are aggressively treated.^{26,90,138} These findings taken collectively suggest therapeutic strategies for lowering BPV-associated dementia risk should focus more specifically on lowering BPV, in addition to treatments that aggressively lower average blood pressure.

CONCLUSIONS



High BPV is a predictor of early AD pathophysiological changes across multiple research groups and older adult cohorts. Specifically, higher BPV is most consistently associated with tau pathology. These relationships are independent of average blood pressure, suggesting that elevated BPV is a distinct vascular risk factor for AD pathophysiology.

Several potential mechanisms for BPV adverse effects related to AD exist, with cerebral hypoperfusion currently having the most support, but more research is needed into the mechanisms linking BPV to AD pathophysiology.

Specific treatments to address elevated BPV are critically needed. Some antihypertensive medications are associated with both decreased BPV and decreased dementia risk, but targeted research is needed to advance therapeutic strategies for lowering BPV beyond current approaches designed to lower average blood pressure alone.

ARTICLE INFORMATION

Affiliations

Leonard Davis School of Gerontology (T.L., D.A.N.), Department of Neurology, Keck School of Medicine (T.L.), Viterbi School of Engineering (N.M.P.), Michelson Institute of Convergent Bioscience (N.M.P.), and Keck School of Medicine (D.A.N.), University of Southern California. Department of Neurology, University of California San Francisco School of Medicine (I.S.). Department of Epidemiology, Harvard T. H. Chan School of Public Health (Y.M.).

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Disclosures

None.

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